

Лабораторная работа №7

Архитектура компьютера

Докладчик

- Аль-хадри Мохаммед Фуад Мохаммед Хамуд
- НКАбд-01-25
- Российский университет дружбы народов им. П. Лумумбы
- 1132255653

Цель работы

Ознакомление с файловой системой Linux, её структурой и основными принципами работы. Получение практических навыков по использованию основных команд Linux для работы с файлами и каталогами, изменению прав доступа, а также анализу состояния файловой системы.

Файловая система Linux представляет собой иерархическую структуру каталогов и файлов. Каждый файл или каталог имеет своё имя, расположение и набор прав доступа.

Основные команды

touch
cat
less
head
tail
cp
mv
mkdir
chmod
df
fsck
mount
kill

Шаг 1

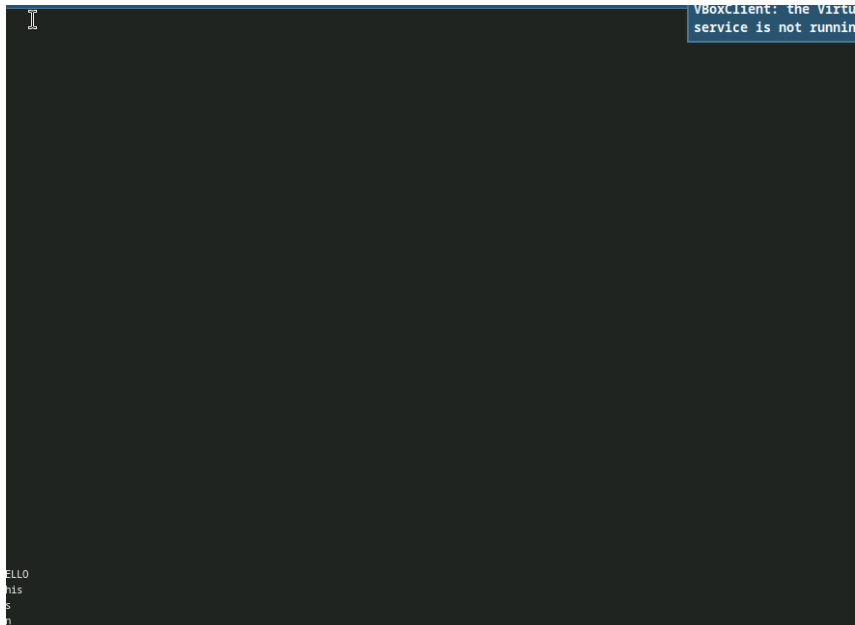
```
aljkhadrimokhl@mkaljkhadrimokhl:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$ touch example.txt
aljkhadrimokhl@mkaljkhadrimokhl:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$ ls
example.txt  text.txt
aljkhadrimokhl@mkaljkhadrimokhl:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$
```

Шаг 2

```
mkaljkhadrimokhl@mkaljkhadrimokhl:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$ nano example.txt
mkaljkhadrimokhl@mkaljkhadrimokhl:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$ cat example.txt
HELLO
this
is
an
example
file

mkaljkhadrimokhl@mkaljkhadrimokhl:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$
```

Шаг 3



Шаг 4

```
mkaljkhadrimokh1@mkaljkhadrimokh1:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$ head [-3] example.txt
head: невозможно открыть '[-3]' для чтения: Нет такого файла или каталога
==> example.txt <==
HELLO
this
is
an
example
file

mkaljkhadrimokh1@mkaljkhadrimokh1:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$ tail [-2] example.txt
tail: невозможно открыть '[-2]' для чтения: Нет такого файла или каталога
==> example.txt <==
HELLO
this
is
an
example
file

mkaljkhadrimokh1@mkaljkhadrimokh1:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$ s
```

Шаг 5

```
mkaljkhadrimokhl1@mkaljkhadrimokhl1:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$ cp example.txt examplr_copy.txt
mkaljkhadrimokhl1@mkaljkhadrimokhl1:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$ ls
example.txt  examplr_copy.txt  text.txt
mkaljkhadrimokhl1@mkaljkhadrimokhl1:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$ cat examplr_copy.txt
HELLO
his
s
n
example
file

mkaljkhadrimokhl1@mkaljkhadrimokhl1:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$ S
```

Шаг 6

```
mkaljkhadrimokhl@mkaljkhadrimokhl:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$ mkdir examples
mkaljkhadrimokhl@mkaljkhadrimokhl:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$ cp example
examples/    example.txt
mkaljkhadrimokhl@mkaljkhadrimokhl:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$ ls
examples  example.txt  examplr_copy.txt  text.txt
mkaljkhadrimokhl@mkaljkhadrimokhl:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$ cp example.txt examplr_copy.txt examples
mkaljkhadrimokhl@mkaljkhadrimokhl:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$ cd examples/
mkaljkhadrimokhl@mkaljkhadrimokhl:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07/examples$ LS
bash: LS: command not found
mkaljkhadrimokhl@mkaljkhadrimokhl:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07/examples$ Ls
bash: Ls: command not found
mkaljkhadrimokhl@mkaljkhadrimokhl:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07/examples$ ls
example.txt  examplr_copy.txt
mkaljkhadrimokhl@mkaljkhadrimokhl:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07/examples$ s
```

Шаг 7

```
mkaljkhadrimokhl@mkaljkhadrimokhl:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07/examples$ cd ..
mkaljkhadrimokhl@mkaljkhadrimokhl:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$ cp examples/example.txt examples/example_copy2
mkaljkhadrimokhl@mkaljkhadrimokhl:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$ ls examples
example_copy2  example.txt  examplr_copy.txt
mkaljkhadrimokhl@mkaljkhadrimokhl:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$
```

Шаг 8

```
example_copy2 example.txt example_copy.txt
mkaljkhadrimokhi@mkaljkhadrimokhi:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$ cp -r examples examples_copy
mkaljkhadrimokhi@mkaljkhadrimokhi:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$ ls
examples examples_copy example.txt example_copy.txt text.txt
mkaljkhadrimokhi@mkaljkhadrimokhi:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$ ls examples_copy/
example_copy2 example.txt example_copy.txt
mkaljkhadrimokhi@mkaljkhadrimokhi:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$
```

Шаг 9

```
mkaljkhadrimokhl@mkaljkhadrimokhl:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$ ls
examples  examples_copy  example.txt  examplr_copy.txt  text.txt
mkaljkhadrimokhl@mkaljkhadrimokhl:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$ mv examplr_copy.txt example_cp.txt
mkaljkhadrimokhl@mkaljkhadrimokhl:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$ ls
example_cp.txt  examples  examples_copy  example.txt  text.txt
mkaljkhadrimokhl@mkaljkhadrimokhl:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$
```

Шаг 10

```
mkaljkhadrimokhl@mkaljkhadrimokhl:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$ mkdir new_folder
mkaljkhadrimokhl@mkaljkhadrimokhl:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$ mv example_cp.txt new_folder/
mkaljkhadrimokhl@mkaljkhadrimokhl:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$ ls
examples  examples_copy  example.txt  new_folder  text.txt
mkaljkhadrimokhl@mkaljkhadrimokhl:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$ ls new_folder
example_cp.txt
mkaljkhadrimokhl@mkaljkhadrimokhl:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$
```

Шаг 11

```
example_cp.txt
kaljkhadrimokhi@kaljkhadrimokhi:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$ mv examples new_folder
kaljkhadrimokhi@kaljkhadrimokhi:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$ ls
examples_copy  example.txt  new_folder  text.txt
kaljkhadrimokhi@kaljkhadrimokhi:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$ ls new_folder/
example_cp.txt  examples
```


Шаг 12

```
mkaljkhadrimokhl@mkaljkhadrimokhl:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$ examples_copy/examplr_copy.txt new_folder/  
bash: examples_copy/examplr_copy.txt: Отказано в доступе  
mkaljkhadrimokhl@mkaljkhadrimokhl:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$
```

Шаг 13

```
drwxr-xr-x. 1 mkaljkhadrimokhl mkaljkhadrimokhl 80 map 28 00:03 examples_copy
-rw-r--r--. 1 mkaljkhadrimokhl mkaljkhadrimokhl 37 map 27 23:46 example.txt
drwxr-xr-x. 1 mkaljkhadrimokhl mkaljkhadrimokhl 44 map 28 00:32 new_folder
-rw-r--r--. 1 mkaljkhadrimokhl mkaljkhadrimokhl 6 map 27 22:51 text.txt
mkaljkhadrimokhl@mkaljkhadrimokhl: ~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$ ls -l
итого 8
mkaljkhadrimokhl@mkaljkhadrimokhl: ~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$
```

Шаг 14

```
mkaljkhadrimokh1@mkaljkhadrimokh1:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$ ls -l
итого 8
drwxr-xr-x. 1 mkaljkhadrimokh1 mkaljkhadrimokh1 48 map 28 00:44 examples_copy
-rw-r--r--. 1 mkaljkhadrimokh1 mkaljkhadrimokh1 37 map 27 23:46 example.txt
drwxr-xr-x. 1 mkaljkhadrimokh1 mkaljkhadrimokh1 76 map 28 00:44 new_folder
-rw-r--r--. 1 mkaljkhadrimokh1 mkaljkhadrimokh1 6 map 27 22:51 text.txt
mkaljkhadrimokh1@mkaljkhadrimokh1:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$ chmod u+x example.txt
mkaljkhadrimokh1@mkaljkhadrimokh1:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$ ls -l
итого 8
drwxr-xr-x. 1 mkaljkhadrimokh1 mkaljkhadrimokh1 48 map 28 00:44 examples_copy
-rwxr--r--. 1 mkaljkhadrimokh1 mkaljkhadrimokh1 37 map 27 23:46 example.txt
drwxr-xr-x. 1 mkaljkhadrimokh1 mkaljkhadrimokh1 76 map 28 00:44 new_folder
-rw-r--r--. 1 mkaljkhadrimokh1 mkaljkhadrimokh1 6 map 27 22:51 text.txt
mkaljkhadrimokh1@mkaljkhadrimokh1:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$
```

Шаг 15

```
mkaljkhadrimokhl@mkaljkhadrimokhl:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$ ls -l
итого 8
drwxr-xr-x. 1 mkaljkhadrimokhl mkaljkhadrimokhl 48 map 28 00:44 examples_copy
-rwxr--r--. 1 mkaljkhadrimokhl mkaljkhadrimokhl 37 map 27 23:46 example.txt
drwxr-xr-x. 1 mkaljkhadrimokhl mkaljkhadrimokhl 76 map 28 00:44 new_folder
-rw-r--r--. 1 mkaljkhadrimokhl mkaljkhadrimokhl 6 map 27 22:51 text.txt
mkaljkhadrimokhl@mkaljkhadrimokhl:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$ chmod o-r example.txt
mkaljkhadrimokhl@mkaljkhadrimokhl:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$ ls -l
итого 8
drwxr-xr-x. 1 mkaljkhadrimokhl mkaljkhadrimokhl 48 map 28 00:44 examples_copy
-rwxr-----. 1 mkaljkhadrimokhl mkaljkhadrimokhl 37 map 27 23:46 example.txt
drwxr-xr-x. 1 mkaljkhadrimokhl mkaljkhadrimokhl 76 map 28 00:44 new_folder
-rw-r--r--. 1 mkaljkhadrimokhl mkaljkhadrimokhl 6 map 27 22:51 text.txt
mkaljkhadrimokhl@mkaljkhadrimokhl:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$
```

Шаг 16

```
mkaljkhadrिमokhl@mkaljkhadrिमokhl:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$ ls -l
итого 8
drwxr-xr-x. 1 mkaljkhadrिमokhl mkaljkhadrिमokhl 48 map 28 00:44 examples_copy
-rwxr-----. 1 mkaljkhadrिमokhl mkaljkhadrिमokhl 37 map 27 23:46 example.txt
drwxr-xr-x. 1 mkaljkhadrिमokhl mkaljkhadrिमokhl 76 map 28 00:44 new_folder
-rw-r--r--. 1 mkaljkhadrिमokhl mkaljkhadrिमokhl 6 map 27 22:51 text.txt
mkaljkhadrिमokhl@mkaljkhadrिमokhl:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$ chmod g+w new_folder
mkaljkhadrिमokhl@mkaljkhadrिमokhl:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$ ls -l
итого 8
drwxr-xr-x. 1 mkaljkhadrिमokhl mkaljkhadrिमokhl 48 map 28 00:44 examples_copy
-rwxr-----. 1 mkaljkhadrिमokhl mkaljkhadrिमokhl 37 map 27 23:46 example.txt
drwxrwxr-x. 1 mkaljkhadrिमokhl mkaljkhadrिमokhl 76 map 28 00:44 new_folder
-rw-r--r--. 1 mkaljkhadrिमokhl mkaljkhadrिमokhl 6 map 27 22:51 text.txt
mkaljkhadrिमokhl@mkaljkhadrिमokhl:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$
```

```

mkal@khadrimokhl:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$ mount
/dev/sda3 on / type btrfs (rw,relatime,seclabel,compress=zstd:1,space_cache=v2,subvol=258,subvol=/root)
devtmpfs on /dev type devtmpfs (rw,nosuid,seclabel,size=4035904k,nr_inodes=1008976,mode=755,inode64)
tmpfs on /dev/shm type tmpfs (rw,nosuid,nodev,seclabel,inode64,usrquota)
devpts on /dev/pts type devpts (rw,nosuid,noexec,relatime,seclabel,gid=5,mode=600,ptmxmode=000)
sysfs on /sys type sysfs (rw,nosuid,nodev,noexec,relatime,seclabel)
securityfs on /sys/kernel/security type securityfs (rw,nosuid,nodev,noexec,relatime)
cgroup2 on /sys/fs/cgroup type cgroup2 (rw,nosuid,nodev,noexec,relatime,seclabel,nsdelegate,memory_recursiveprot)
none on /sys/fs/pstore type pstore (rw,nosuid,nodev,noexec,relatime,seclabel)
bpf on /sys/fs/bpf type bpf (rw,nosuid,nodev,noexec,relatime,mode=700)
configfs on /sys/kernel/config type configfs (rw,nosuid,nodev,noexec,relatime)
proc on /proc type proc (rw,nosuid,nodev,noexec,relatime)
tmpfs on /run type tmpfs (rw,nosuid,nodev,seclabel,size=1623624k,nr_inodes=819200,mode=755,inode64)
selinuxfs on /sys/fs/selinux type selinuxfs (rw,nosuid,noexec,relatime)
systemd-1 on /proc/sys/fs/binfmt_misc type autofs (rw,relatime,fd=42,pgpr=1,timeout=0,minproto=5,maxproto=5,direct,pipe_ino=3003)
tracefs on /sys/kernel/tracing type tracefs (rw,nosuid,nodev,noexec,relatime,seclabel)
mqueue on /dev/mqueue type mqueue (rw,nosuid,nodev,noexec,relatime,seclabel)
debugfs on /sys/kernel/debug type debugfs (rw,nosuid,nodev,noexec,relatime,seclabel)
hugetlbfs on /dev/hugepages type hugetlbfs (rw,nosuid,nodev,relatime,seclabel,pagesize=2M)
fusectl on /sys/fs/fuse/connections type fusectl (rw,nosuid,nodev,noexec,relatime)
tmpfs on /run/credentials/systemd-journald.service type tmpfs (ro,nosuid,nodev,noexec,relatime,nosymfollow,seclabel,size=1024k,nr_inodes=1024,mode=700,inode64,noswap)
tmpfs on /tmp type tmpfs (rw,nosuid,nodev,seclabel,nr_inodes=1048576,inode64,usrquota)
/dev/sda3 on /home type btrfs (rw,relatime,seclabel,compress=zstd:1,space_cache=v2,subvol=257,subvol=/home)
/dev/sda2 on /boot type ext4 (rw,relatime,seclabel)
tmpfs on /run/credentials/systemd-resolved.service type tmpfs (ro,nosuid,nodev,noexec,relatime,nosymfollow,seclabel,size=1024k,nr_inodes=1024,mode=700,inode64,noswap)
sunrpc on /var/lib/nfs/rpc_pipefs type rpc_pipefs (rw,relatime)
tmpfs on /run/user/1000 type tmpfs (rw,nosuid,nodev,relatime,seclabel,size=811808k,nr_inodes=202952,mode=700,uid=1000,gid=1000,inode64)
portal on /run/user/1000/doc type fuse.portal (rw,nosuid,nodev,relatime,user_id=1000,group_id=1000)
tmpfs on /run/user/0 type tmpfs (rw,nosuid,nodev,relatime,seclabel,size=811808k,nr_inodes=202952,mode=700,inode64)
mkal@khadrimokhl:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$

```

Шаг 18

```
mkaljkhadrimokhl1@mkaljkhadrimokhl1:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$ cat /etc/fstab
#
# /etc/fstab
# Created by anaconda on Wed Mar 18 02:25:10 2026
#
# Accessible filesystems, by reference, are maintained under '/dev/disk/'.
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.
#
# After editing this file, run 'systemctl daemon-reload' to update systemd
# units generated from this file.
#
UUID=7a320771-96b0-430f-9259-b2f0277c8fe1 / btrfs subvol=root,compress=zstd:1 0 0
UUID=7e6f5c8b-0c37-4237-b1ee-9ca929ebccc1 /boot ext4 defaults 1 2
UUID=7a320771-96b0-430f-9259-b2f0277c8fe1 /home btrfs subvol=home,compress=zstd:1 0 0
mkaljkhadrimokhl1@mkaljkhadrimokhl1:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$
```

```
mkaljkhadrimokhl1@mkaljkhadrimokhl1:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$ df
Файловая система 1К-блоков  Иستخدمانو  Доступно  Иستخدمانو%  Смонтировано в
/dev/sda3          92271616      3607932  86869044          4% /
devtmpfs           4035904         0  4035904          0% /dev
tmpfs              4059056      3580  4055476          1% /dev/shm
tmpfs              1623624      1140  1622484          1% /run
tmpfs              1024         0    1024          0% /run/credentials/systemd-journald.service
tmpfs              4059056         4  4059052          1% /tmp
/dev/sda3          92271616      3607932  86869044          4% /home
/dev/sda2          1992552      448220  1423092         24% /boot
tmpfs              1024         0    1024          0% /run/credentials/systemd-resolved.service
tmpfs              811808         96  811712          1% /run/user/1000
tmpfs              811808         60  811748          1% /run/user/0
mkaljkhadrimokhl1@mkaljkhadrimokhl1:~/Study/Workspaces/study_2025-2026_os-intro/labs/lab07$
```



```
mkaljkhadrimokhl@mkaljkhadrimokhl1:/usr/include/sys$ ls -l io.h
-rw-r--r--. 1 root root 5086 фев 18 03:00 io.h
mkaljkhadrimokhl@mkaljkhadrimokhl1:/usr/include/sys$ chmod u+x g+wx
chmod: невозможно получить доступ к 'g+wx': Нет такого файла или каталога
mkaljkhadrimokhl@mkaljkhadrimokhl1:/usr/include/sys$ chmod u+x io.h
chmod: изменение прав доступа для 'io.h': Операция не позволена
mkaljkhadrimokhl@mkaljkhadrimokhl1:/usr/include/sys$ sudo chmod u+x io.h
[sudo] пароль для mkaljkhadrimokhl1:
mkaljkhadrimokhl@mkaljkhadrimokhl1:/usr/include/sys$ sudo chmod g+wx o.h
chmod: невозможно получить доступ к 'o.h': Нет такого файла или каталога
mkaljkhadrimokhl@mkaljkhadrimokhl1:/usr/include/sys$ sudo chmod g+wx io.h
mkaljkhadrimokhl@mkaljkhadrimokhl1:/usr/include/sys$ ls -l io.h
-rwxrwxr--. 1 root root 5086 фев 18 03:00 io.h
mkaljkhadrimokhl@mkaljkhadrimokhl1:/usr/include/sys$ S
```

Шаг 21

```
mkaljkhadrimokhl@mkaljkhadrimokhl:/$ sudo cp /usr/include/sys/io.h /home
mkaljkhadrimokhl@mkaljkhadrimokhl:/$ ls
afs bin boot dev etc home lib lib64 lost+found media mnt opt proc root run sbin srv sys tmp usr var
mkaljkhadrimokhl@mkaljkhadrimokhl:/$ cd home
mkaljkhadrimokhl@mkaljkhadrimokhl:/home$ ls
io.h mkaljkhadrimokhl
mkaljkhadrimokhl@mkaljkhadrimokhl:/home$
```

Шаг 22

```
mkaljkhadrimokhl@mkaljkhadrimokhl:/usr/include/sys$ cp /usr/include/sys/io.h /home  
mkaljkhadrimokhl:/home$ cp: невозможно создать обычный файл '/home/io.h': Отказано в доступе  
mkaljkhadrimokhl@mkaljkhadrimokhl:/usr/include/sys$ cd /
```

Шаг 23

```
mkaljkhadrimokhl@mkaljkhadrimokhl:~$ sudo cp /usr/include/sys/io.h /home/equipment
mkaljkhadrimokhl@mkaljkhadrimokhl:~$ ls
equipment  io.h  mkaljkhadrimokhl
mkaljkhadrimokhl@mkaljkhadrimokhl:~$
```

Шаг 24

```
mkaljkhadrimokhl@mkaljkhadrimokhl:/$ mkdir ~/ski.places  
mkdir: невозможно создать каталог «/home/mkaljkhadrimokhl/ski.places»: Файл существует  
mkaljkhadrimokhl@mkaljkhadrimokhl:/$
```

Шаг 25

```
mkaljkhadrimokhl@mkaljkhadrimokhl:/$ mv /home/equipment ~/ski.spaces/
mv: невозможно переместить '/home/equipment' в '/home/mkaljkhadrimokhl/ski.spaces/equipment': Отказано в доступе
mkaljkhadrimokhl@mkaljkhadrimokhl:/$ sudo mv /home/equipment ~/ski.spaces/
mkaljkhadrimokhl@mkaljkhadrimokhl:/$ ls ~/
ski.spaces  Study  Видео  Документы  Загрузки  Изображения  Музыка  Общедоступные  'Рабочий стол'  Шаблоны
mkaljkhadrimokhl@mkaljkhadrimokhl:/$ ls ~/ski.spaces/
equipment
mkaljkhadrimokhl@mkaljkhadrimokhl:/$
```

Шаг 26

```
mkaljkhadrimokhl@mkaljkhadrimokhl:/$ mv ~/ski.spaces/equipment ~/ski.spaces/equiplist
mkaljkhadrimokhl@mkaljkhadrimokhl:/$ mv ~/ski.spaces/equipment ~/ski.plases/equiplist
mv: не удалось выполнить stat для '/home/mkaljkhadrimokhl/ski.spaces/equipment': Нет такого файла или каталога
mkaljkhadrimokhl@mkaljkhadrimokhl:/$ mv ~/ski.spaces ~/ski.plases
mkaljkhadrimokhl@mkaljkhadrimokhl:/$ ls ~/
ski.plases  Study  Видео  Документы  Загрузки  Изображения  Музыка  Общедоступные  'Рабочий стол'  Шаблоны
mkaljkhadrimokhl@mkaljkhadrimokhl:/$ ls ~/ski.plases/
equiplist
mkaljkhadrimokhl@mkaljkhadrimokhl:/$
```

```
mkaljkhadrimokhl@mkaljkhadrimokhl:/$ ls ~/ski.plases/  
equiplist  
mkaljkhadrimokhl@mkaljkhadrimokhl:/$ touch ~/abc1; cp ~/abc1 ~/ski.plases/equiplist2  
mkaljkhadrimokhl@mkaljkhadrimokhl:/$ ls ~/ski.plases/  
equiplist  equiplist2  
mkaljkhadrimokhl@mkaljkhadrimokhl:/$
```


Шаг 28

```
mkaljkhadrimokhl@mkaljkhadrimokhl:/$ mkdir ~/ski.plases/equipment
mkdir: невозможно создать каталог «/home/mkaljkhadrimokhl/ski.plases/equipment»: Файл существует
mkaljkhadrimokhl@mkaljkhadrimokhl:/$ ls ~/ski.plases
equiplist  equiplist2  equipment
mkaljkhadrimokhl@mkaljkhadrimokhl:/$
```

```
mkaljkhadrimokhl@mkaljkhadrimokhl:/$ ls ~/ski.places
equiplist  equiplist2  equipment
mkaljkhadrimokhl@mkaljkhadrimokhl:/$ mv ~/ski.places/equiplist ~/ski.places/equiplist2 ~/ski.places/equipment
mkaljkhadrimokhl@mkaljkhadrimokhl:/$ ls ~/ski.places
ls: невозможно получить доступ к '/home/mkaljkhadrimokhl/ski.places': Нет такого файла или каталога
mkaljkhadrimokhl@mkaljkhadrimokhl:/$ ls ~/ski.places
equipment
mkaljkhadrimokhl@mkaljkhadrimokhl:/$ ls ~/ski.places/equipment/
equiplist  equiplist2
mkaljkhadrimokhl@mkaljkhadrimokhl:/$
```

Шаг 30

```
mkaljkhadrimokhl@mkaljkhadrimokhl:/$ ls ~/ski.places/equipment/  
equiplist equiplist2  
mkaljkhadrimokhl@mkaljkhadrimokhl:/$ mkdir ~/newdir; mv ~/newdir ~/ski.places/plans  
mkaljkhadrimokhl@mkaljkhadrimokhl:/$ ls ~/ski.places/  
equipment plans  
mkaljkhadrimokhl@mkaljkhadrimokhl:/$
```

```
mkaljkhadrimokhl@mkaljkhadrimokhl:/$ mkdir ~/ski.places/chmod_files
mkaljkhadrimokhl@mkaljkhadrimokhl:/$ cd ~/ski.places/chmod_files
mkaljkhadrimokhl@mkaljkhadrimokhl:~/ski.places/chmod_files$ mkdir australia play; touch my_os feathers
mkaljkhadrimokhl@mkaljkhadrimokhl:~/ski.places/chmod_files$ ls
australia  feathers  my_os  play
mkaljkhadrimokhl@mkaljkhadrimokhl:~/ski.places/chmod_files$ ls -l
итого 0
drwxr-xr-x. 1 mkaljkhadrimokhl mkaljkhadrimokhl 0 map 28 17:39 australia
-rw-r--r--. 1 mkaljkhadrimokhl mkaljkhadrimokhl 0 map 28 17:39 feathers
-rw-r--r--. 1 mkaljkhadrimokhl mkaljkhadrimokhl 0 map 28 17:39 my_os
drwxr-xr-x. 1 mkaljkhadrimokhl mkaljkhadrimokhl 0 map 28 17:39 play
mkaljkhadrimokhl@mkaljkhadrimokhl:~/ski.places/chmod_files$ S
```

```
root
mkaljkhadrimokh1@mkaljkhadrimokh1:~/ski.places/chmod_files$ chmod o-x australia/
mkaljkhadrimokh1@mkaljkhadrimokh1:~/ski.places/chmod_files$ ls -l
итого 0
drwxr--r--. 1 mkaljkhadrimokh1 mkaljkhadrimokh1 0 map 28 17:39 australia
-rw-r--r--. 1 mkaljkhadrimokh1 mkaljkhadrimokh1 0 map 28 17:39 feathers
-rw-r--r--. 1 mkaljkhadrimokh1 mkaljkhadrimokh1 0 map 28 17:39 my_os
drwxr-xr-x. 1 mkaljkhadrimokh1 mkaljkhadrimokh1 0 map 28 17:39 play
```

```
mkaljkhadrimokh1@mkaljkhadrimokh1:~/ski.plases/chmod_files$ chmod g-r play
mkaljkhadrimokh1@mkaljkhadrimokh1:~/ski.plases/chmod_files$ chmod g+x play/
mkaljkhadrimokh1@mkaljkhadrimokh1:~/ski.plases/chmod_files$ ls -l
того 0
-rwxr--r--. 1 mkaljkhadrimokh1 mkaljkhadrimokh1 0 map 28 17:39 australia
-rw-r--r--. 1 mkaljkhadrimokh1 mkaljkhadrimokh1 0 map 28 17:39 feathers
-rw-r--r--. 1 mkaljkhadrimokh1 mkaljkhadrimokh1 0 map 28 17:39 my_os
-rwx--xr-x. 1 mkaljkhadrimokh1 mkaljkhadrimokh1 0 map 28 17:39 play
mkaljkhadrimokh1@mkaljkhadrimokh1:~/ski.plases/chmod_files$ chmod o-r play/
mkaljkhadrimokh1@mkaljkhadrimokh1:~/ski.plases/chmod_files$ ls -l
того 0
-rwxr--r--. 1 mkaljkhadrimokh1 mkaljkhadrimokh1 0 map 28 17:39 australia
-rw-r--r--. 1 mkaljkhadrimokh1 mkaljkhadrimokh1 0 map 28 17:39 feathers
-rw-r--r--. 1 mkaljkhadrimokh1 mkaljkhadrimokh1 0 map 28 17:39 my_os
-rwx--x--x. 1 mkaljkhadrimokh1 mkaljkhadrimokh1 0 map 28 17:39 play
mkaljkhadrimokh1@mkaljkhadrimokh1:~/ski.plases/chmod_files$ S
```

Шаг 34

```
то не каталог
ljkhadrimokhi:~/ski_places/chmod_files$ cat /etc/passwd
x:/root:/bin/bash
usz/sbin/nologin
/sbin:usz/sbin/nologin
```

```
mkaljkhadrimokhl@mkaljkhadrimokhl:~$ sudo mount /dev/cdrom /mnt
[sudo] пароль для mkaljkhadrimokhl:
mount: /mnt: WARNING: source write-protected, mounted read-only.
mkaljkhadrimokhl@mkaljkhadrimokhl:~$ sudo /mnt/VBoxLinuxAdditions.run
Verifying archive integrity... 100% MD5 checksums are OK. All good.
Uncompressing VirtualBox 7.2.6 Guest Additions for Linux 100%
VirtualBox Guest Additions installer
Removing installed version 7.2.6 of VirtualBox Guest Additions...
VirtualBox Guest Additions: Starting.
VirtualBox Guest Additions: Setting up modules
VirtualBox Guest Additions: Building the VirtualBox Guest Additions kernel
modules. This may take a while.
VirtualBox Guest Additions: To build modules for other installed kernels, run
VirtualBox Guest Additions: /sbin/rcvboxadd quicksetup <version>
VirtualBox Guest Additions: or
VirtualBox Guest Additions: /sbin/rcvboxadd quicksetup all
VirtualBox Guest Additions: Building the modules for kernel
6.19.8-200.fc43.x86_64.

VirtualBox Guest Additions: Look at /var/log/vboxadd-setup.log to find out what
went wrong
File context for /opt/VBoxGuestAdditions-7.2.6/other/mount.vboxsf already defined, modifying instead
VirtualBox Guest Additions: Running kernel modules will not be replaced until
the system is restarted or 'rcvboxadd reload' triggered
VirtualBox Guest Additions: reloading kernel modules and services
VirtualBox Guest Additions: cannot reload kernel modules: one or more module(s)
is still in use
```


В ходе выполнения лабораторной работы была изучена файловая система Linux и её структура. Были освоены основные команды для работы с файлами и каталогами.